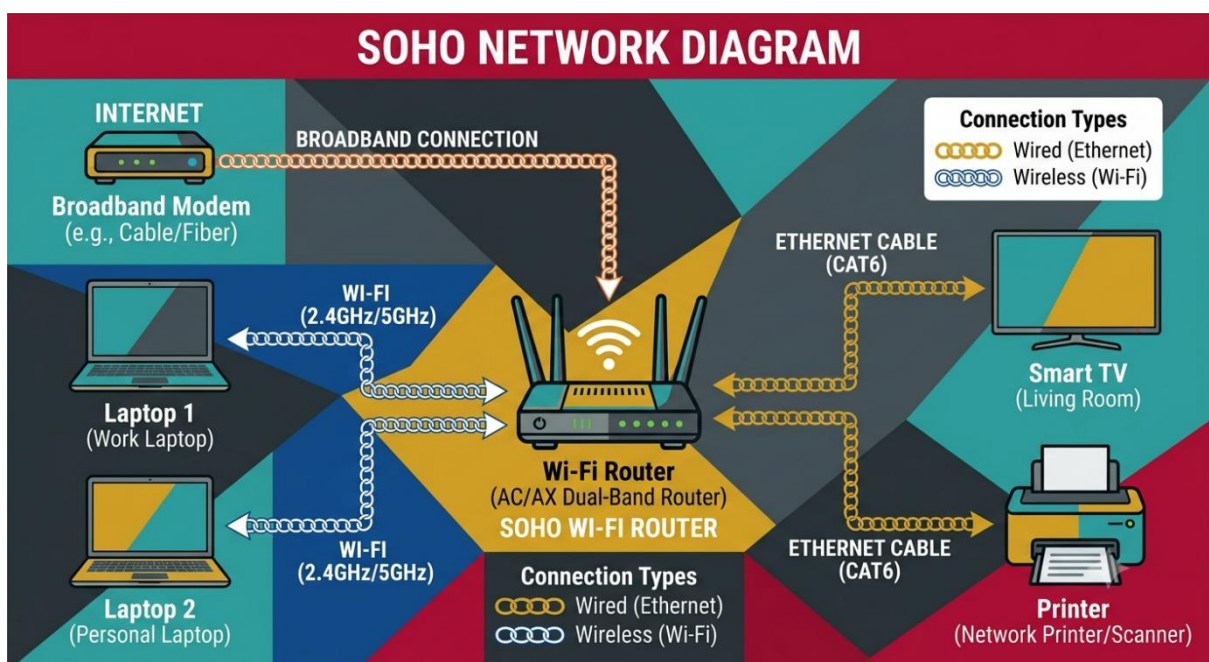


BUILD A SOHO NETWORK

1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

Answer



2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed. Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Answer

A generic modern SOHO checklist often includes items like these (these are only placeholders until the diagram is available):

- ISP modem/ONT (usually provided by the ISP)*
- Dual-band or Wi-Fi 6 router (e.g., TP-Link Archer AX series, D-Link DIR, Netgear, or ISP-provided)*
- Gigabit switch (if >4-5 wired devices)*

- *Ethernet cables (Cat6)*
- *Optional mesh nodes or access points for coverage*
- *PoE injector/switch if cameras or APs need power over Ethernet*

Here are high-resolution views of each SOHO networking component to give you a clear look at their physical ports, build, and form factor:

TP-Link Archer C6 AC1200 Router

- *Design Features: Features a textured diamond chassis with 4 external high-gain antennas for wide wireless coverage and 1 internal antenna. The back houses 1 Gigabit WAN port, 4 Gigabit LAN ports, a WPS/Wi-Fi button, and the power switch.*



TP-Link TL-SG108 8-Port Desktop Switch

- *Design Features: Compact, fanless steel enclosure built for quiet desktop placement or wall mounting. Features 8*

auto-negotiating Gigabit RJ-45 ports with LED activity indicators above each port.



UGREEN Cat 6 Gigabit Ethernet Cable

- Design Features: Equipped with gold-plated 8P8C RJ45 connectors, molded strain-relief boots, and pure copper conductors designed to prevent cross-talk and maintain full 1000 Mbps gigabit throughput.*



Component	Function	Suggested Model	Approx. Price (INR)
ISP Modem / ONU	Connects to ISP (Fibre/Broadband)	Provided by ISP (e.g., Airtel / JioFiber)	Included with connection

Component	Function	Suggested Model	Approx. Price (INR)
Main Wi-Fi Router	Handles Routing, DHCP, NAT, & Wireless	TP-Link Archer C6	₹2,500 – ₹2,800
Network Switch	Expands wired LAN ports	TP-Link TL-SG108	₹1,700 – ₹2,200
Access Point / Extender	Extends wireless coverage	TP-Link Archer C6 or TP-Link RE305	₹2,000 – ₹2,700
Ethernet Cables	Interconnects all network devices	UGREEN Cat6 Cable or FEDUS Cat6	₹150 – ₹500 (per cable)

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Answer

Router configuration steps

1. *Connect the ISP modem/ONT to the router's WAN/Internet port.*

2. *Connect your computer or phone to the router using Wi-Fi or a LAN cable.*
3. *Open a browser and visit 192.168.1.1, 192.168.0.1, or the address printed on the router.*
4. *Log in using the default details printed on the router label.*
5. *Open Wireless/Wi-Fi Settings.*
6. *Change the SSID to a unique name, such as Manan_SOHO.*
7. *Select WPA2-PSK/AES security. Avoid WEP or open security.*
8. *Create a strong Wi-Fi password with at least 12 characters.*
9. *Open Administration/System/Management settings.*
10. *Change the default admin password to a different strong password.*
11. *Click Save/Apply and allow the router to restart.*
12. *Reconnect all devices using the new SSID and Wi-Fi password.*
13. *Check that Internet access works by opening a website or using ping 8.8.8.8.*

4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Answer

SOHO Network IP Configuration

Network settings

- *Router/Gateway: 192.168.1.1*
- *Subnet mask: 255.255.255.0*
- *DHCP range: 192.168.1.100 – 192.168.1.200*

Device	MAC Address	IP Address	Assignment
Router	00:11:22:33:44:55	192.168.1.1	Static
Printer	A1:B2:C3:D4:E5:F6	192.168.1.10	Static
Smart TV	11:22:33:44:55:66	192.168.1.20	Static
Laptop 1	AA:BB:CC:DD:EE:01	192.168.1.101	DHCP
Laptop 2	AA:BB:CC:DD:EE:02	192.168.1.102	DHCP

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Answer

1.Fix Invalid IP Address (DHCP Failure):1 min.

Open Command Prompt and run ipconfig /release then ipconfig /renew. Ensure TCP/IP settings are set to Obtain an IP address automatically.

2.Fix Physical or Wi-Fi Connection:2 min.

If wired, swap the Ethernet cable or try a different port on the switch. If on Wi-Fi, turn Airplane Mode off, Forget the Wi-Fi network, and reconnect with the password.

3.Fix DNS Failure:2 min.

Run ipconfig /flushdns and change your adapter DNS settings manually to Cloudflare (1.1.1.1) or Google (8.8.8.8).

- ***How to Verify:** Open Command Prompt and run ping 8.8.8.8 followed by ping google.com. Receiving reply packets on both confirms the issue is fixed.*

